Paradigma Zero: A Dynamically Consistent Scenario for Gravitational Capture of the Moon — Computational Evidence and Observational Convergence from 32 Independent Datasets

Alexssandro A.
Independent Researcher, Osasco, SP, Brazil
ALEX7NDO@GMAIL.COM

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Abstract

The origin of the Moon remains a central unsolved problem in planetary science. Four high-precision independent measurements present collectively unresolved anomalies within the canonical Giant Impact model: the oxygen isotope identity between Earth and Moon indistinguishable at sub-ppm precision ($\Delta\Delta'^{17}\mathrm{O} = 0.2 \pm 1.6$ ppm; Fischer et al., 2024); mantle thermal asymmetry between the nearside and farside of 100–200 K (Park et al., 2025; He et al., 2025); the Lunar Laser Ranging kinematic paradox, in which the measured recession rate ($da/dt = 38.08 \pm 0.19$ mm yr^{-1}; Williams et al., 2014), linearly extrapolated, implies a lunar origin ∼10 Gyr ago, exceeding the age of the Solar System; and the anomalously intense primordial lunar dynamo generating fields of 20–110 μT between 4.25 and 3.56 Gyr by a mechanism that remains unexplained (Weiss & Tikoo, 2014).

We report the results of a systematic, unguided parameter-space exploration comprising ∼65,000 N-body integrations across 26 simulation stages, executed with REBOUND/IAS15 without a predetermined capture scenario. Configurations incorporating a third gravitational perturber (Body Z) consistently produced dynamical architectures incompatible with the observed Solar System and were rejected by the simulations themselves, without authorial intervention. In the absence of that third body, two discrete canonical configurations emerged: candidate P1-06 ($a = 1.70$ AU, $e = 0.43$,$i = 10°$, $r_{\min} = 15.51\ R_\oplus$, $\Delta T = 17.8$ yr) and candidate P1-03 ($a = 1.65$ AU, $e = 0.40$, $i = 20°$, $r_{\min} = 9.84\ R_\oplus$, $\Delta T = 122.6$ yr). In both, the Moon occupied a heliocentric orbit at 1.65–1.70 AU — while Proto-Mars maintained the 8:5 Earth–Proto-Mars mean-motion resonance at 1.368 AU — and is proposed to have been gravitationally captured by the Earth at approximately 12,000 BP, a chronological inference supported by convergent geological and geophysical evidence (Section 3.5) and identified as the preferred temporal hypothesis

of this work. Capture required tidal energy dissipation during the close encounter, formally necessary per Bondi (1979), who demonstrated that two-body gravitational capture is impossible without dissipation.

Post-hoc evaluation against 32 independent peer-reviewed observational datasets — 30 Grade A definitive, 1 Grade B, CONVERGE PARTIAL (T-08), and 1 computational result (S-01; Grade B for the author's simulation, Grade A for the theoretical mean-motion resonance framework) — produces no contradiction. Five independently verified retrodictions (R1–R5) are documented, including the nearside thermal excess confirmed by GRAIL tidal tomography (Park et al., 2025) and Chang'e-6 basalt petrology (He et al., 2025), both published after completion of the simulations and both independently quantifying a 100–200 K thermal asymmetry between the nearside and farside, consistent with tidal deformation at $r_{\min} \sim 10$–$15\ R_\oplus$.

Both candidates satisfy three independent orbital constraints (M1–M3; simulation stage F5-v6): terrestrial orbital stability ($a_\oplus = 0.9996$–1.0004 AU); Proto-Mars secular migration from 1.368 AU to $a_{\rm PM} = 1.4954$–1.5169 AU via the Δv impulse at capture followed by g_4–g_3 secular resonance with Jupiter in $\tau \sim 1$ Myr — two mechanistically distinct processes acting on different degrees of freedom (semi-major axis and eccentricity, respectively); lunar approach below the gravitational capture threshold ($r_{\min} < 70\ R_\oplus$); and establishment of terrestrial obliquity from 0° to 23.5° via gravitational torque ($\Delta L = 2.378 \times 10^{33}$ kg m^2 s^{-1}), closed analytically at $E_{\rm torque}/E_{\rm tilt} = 1.0005$ for both candidates. The LLR kinematic paradox (M4) is qualitatively resolved by the recent capture hypothesis, which renders long-term linear extrapolations dynamically invalid; quantitative consistency with $E_0 = 12{,}000$ BP is established for P1-03 by the variable-dissipation tidal model SIM-A v2 (19 of 60 configurations; best $t_{a^*} = 12.19$ kyr, ratio 1.02×; Limitation L3 resolved for P1-03; Section 5.4); resolution for P1-06 is identified as future work (Section 9.7).

Two falsifiable predictions are formally registered. Prediction S-06: a seismic and thermal deficit on the farside testable by the dual FSS+LITMS system at Schrödinger Basin (launch 2027), with confirmation criterion of fewer than 10% of deep moonquake events on the farside and falsification criterion of more than 30% farside distribution. Prediction L-07: K-Ar isotopic reset between 12,000 and 13,000 years BP in Apollo samples from potential tsunami impact zones, with U-Pb in zircon remaining unchanged and serving as an independent chronological anchor; the differential thermal susceptibility between the two systems (K-Ar closure temperature ∼300°C versus U-Pb ∼700–900°C) provides a mechanism-specific discriminant.

The convergence between an unguided computational exploration and independent observational constraints documents a scenario that was not constructed to fit the data, but found within them.

Keywords: lunar origin; gravitational capture; N-body simulation; REBOUND; Earth-Moon system; Younger Dryas; obliquity; Proto-Mars; mean-motion resonance; tidal dissipation

1 Introduction

1.1 Current state of the lunar origin problem

The origin of the Moon is a cornerstone of planetary science, with the Giant Impact hypothesis serving as the canonical framework for decades. While this model successfully accounts for the Earth-Moon system's high angular momentum and the Moon's small metallic core, it faces increasing tension with high-precision measurements. Mainstream investigators have explicitly identified significant gaps in the current paradigm. Regarding the lunar magnetic field, Weiss & Tikoo (2014) state that "The mechanisms for sustaining such an intense and long-lived dynamo are uncertain". Geochemical constraints also present challenges; Righter (2019) noted that "The depletion of many volatile elements remains incompletely explained". Recent isotopic analyses have further deepened this isotopic crisis, as Fischer et al. (2024) observed that "The isotopic similarity of Earth and Moon rocks is a significant puzzle in geo- and cosmochemistry, contradicting prevailing Moon formation models". Consequently, the call for a paradigm shift is growing, with Barboni et al. (2025) concluding that "These results underscore the need for a serious re-evaluation of lunar materials".

1.2 Recent data reopening the debate

New observational evidence across multiple disciplines has reintroduced fundamental questions about the Moon's history. Geochemical data from Fischer et al. (2024) reveals a sub-ppm oxygen isotope identity ($\Delta\Delta'^{17}\mathrm{O} = 0.2 \pm 1.6$ ppm), a level of precision that challenges the likelihood of two distinct bodies (Earth and the hypothetical Theia) reaching such homogeneity. Geophysical measurements from the GRAIL mission indicate a significant thermal asymmetry; Park et al. (2025) reported an anomalous Love number $k_3 = 0.0163 \pm 0.0007$, implying persistent lateral heterogeneity in the lunar mantle. This is independently supported by Chang'e-6 petrology, which found the farside mantle to be approximately 100°C cooler than the nearside (He et al., 2025).

Internal structure models by Briaud et al. (2023) have confirmed a solid inner core of 258 ± 40 km, suggesting a complex mantle overturn history. Paleomagnetic data from Chang'e-5 basalts show a persistent, albeit weak, dynamo of $\sim$2–4 μT as recently as 2.0 Ga (Cai et al., 2025), extending the required duration of lunar core convection beyond standard cooling models. Furthermore, Barboni et al. (2025) demonstrated that supposedly pristine lunar clasts show pervasive impact reworking, suggesting a more violent and energetically complex history than previously assumed. Finally, the kinematic paradox identified in Lunar Laser Ranging (LLR) remains unresolved: at the measured recession rate of $da/dt = 38.08 \pm 0.19$ mm/yr (Williams et al., 2014), a linear retroactive extrapolation places the Moon at the Earth's surface approximately 10 Gyr ago, a result inconsistent with the 4.56 Gyr age of the Solar System.

1.3 Investigated question and adopted method

The investigation presented in this work emerged from an empirically-driven parametric exploration rather than from a theoretical challenge to existing models.

The original inquiry focused on a hypothetical additional gravitational perturber, designated as Body Z, that might have redirected the Moon toward its current configuration.

The methodology employed was an unguided parametric exploration utilizing over 65,000 N-body integrations across 26 simulation stages, executed with the IAS15 integrator within the REBOUND framework (detailed in Section 2). By varying parameters such as semi-major axis (a), eccentricity (e), and inclination (i) without a predetermined outcome, the study aimed to find the most parsimonious orbital history. As the parameter space was systematically swept, simulation outcomes incorporating Body Z consistently produced dynamical instabilities inconsistent with the modern Solar System architecture. Body Z was thus progressively rejected by the simulations themselves, not by theoretical assumption.

The question then arose: could gravitational capture emerge from the existing three-body dynamics of the Sun–Earth–Mars system, without requiring an additional perturber? This possibility is formally non-trivial: as demonstrated by Bondi (1979), gravitational capture in a two-body problem is impossible without an energy dissipation mechanism. In the three-body configuration explored here, tidal dissipation at periapsis within the lunar body itself — which, during its heliocentric phase, maintained a global magma ocean with $k_2/Q \gg 1$, rendering it highly susceptible to tidal energy loss — provides the physical mechanism by which a hyperbolic trajectory is converted into a bound orbit (detailed in Section 2.4). A prerequisite dynamical condition identified by the simulations is the pre-existing 8:5 mean-motion resonance between Earth and Proto-Mars, with Proto-Mars at $a = 1.368$ AU and $e = 0.30$ (established by 288 WHFast integrations across the full parameter space; see Section 4.2).

The legitimacy of this question further rests on whether a singular, overlooked dynamical event can reconcile the 32 independent observational datasets detailed in Section 4. Independent stratigraphic evidence provides an empirical anchor for the proposed event: the Younger Dryas Boundary (YDB, ∼12,800 BP) records platinum-group anomalies, meltglass with formation temperatures exceeding 1,700°C, nanodiamond layers, and synchronous megafaunal extinctions across five continents (Firestone et al., 2007; Kennett et al., 2015), for which no causal agent has achieved consensus in 17 years of investigation. The present work proposes that these signatures are consistent with the final gravitational capture of the Moon, with the dynamical simulations anchoring the capture event to approximately 12,000 BP (event E_0; see Section 4.2). The YDB at ∼12,800 BP corresponds to the surface manifestation of the preceding close-approach and circularization phase (event E_1: 12,134–13,862 BP; Section 4.2), which predates final capture by centuries to millennia. The scope and limitations of this approach are addressed in Section 9.

2 Methods and Evidentiary Framework

2.1 N-body integration pipeline

All N-body integrations were executed using the REBOUND open-source framework (Rein & Liu, 2012), version 4.6.0, with the IAS15 adaptive high-order integrator (Rein & Spiegel, 2015) as the primary solver for all capture-phase and 3D simulations. IAS15 was selected for its high-order adaptive timestep and demonstrated near-machine-

precision energy conservation during close encounters, which are the dynamically critical events in the capture scenario. For the long-timescale Mean Motion Resonance screening (Phase F3), the symplectic WHFast integrator (Rein & Tamayo, 2015) was employed with a fixed timestep of 1/20 of the shortest orbital period.

The five-body system modeled in the canonical capture simulations (SIM-A-P1) comprised the Sun, Earth, Proto-Mars, Moon, and Jupiter. Physical parameters were adopted from the International Earth Rotation and Reference Systems Service (IERS 2010) and the International Astronomical Union (IAU 2015): $M_{\text{Earth}} = 3.003 \times 10^{-6}\ M_\odot$, $M_{\text{Moon}} = 3.694 \times 10^{-8}\ M_\odot$, $M_{\text{Proto-Mars}} = 3.213 \times 10^{-7}\ M_\odot$, $M_{\text{Jupiter}} = 9.545 \times 10^{-4}\ M_\odot$. All integrations were performed in units where $G = 4\pi^2\ \text{AU}^3\ M_\odot^{-1}\ \text{yr}^{-2}$. Gravitational softening $\varepsilon = 3 \times 10^{-7}$ AU was applied, smaller than the lunar physical radius and confirmed to not distort dynamics at relevant scales.

Tidal dissipation during the lunar heliocentric phase was modeled using the Equilibrium Tide model of Eggleton, Kiseleva & Hut (1998), with the eccentricity function $f(e)$ from Hut (1981). The young Moon in a global magma ocean state was assigned $k_2 = 0.80$ (fluid limit, Kelvin 1863) and $Q = 2.0$ (dissipative limit for a liquid body), yielding $k_2/Q = 0.400$ — a factor of 1,667 higher than the present solid Moon. This parameter regime is physically justified by adopting the fluid-body limit for the Love number ($k_2 = 0.80$; Kelvin, 1863) and the high-dissipation regime characteristic of a body sustaining a global magma ocean ($Q \approx 2$), whose existence is independently established by the crystallization chronology of the lunar magma ocean (Nemchin et al., 2009). The resulting $k_2/Q = 0.400$ represents an upper bound on tidal dissipation efficiency for a lunar-sized body in the magma ocean state; the sensitivity of the capture outcome to this parameter is discussed in Section 9.6.

2.2 Evidentiary grade classification

All observational data and computational results cited in this work are classified according to a three-tier evidentiary grade system, following the convention of independent verifiability:

Grade A: Established, peer-reviewed observational data derived from direct measurement or well-validated physical models. These data points are independent of the present work and existed prior to or without reference to the Paradigma Zero scenario.

Grade B: Computational results derived from simulations conducted in this work. These are not independent of the scenario but constitute internally consistent quantitative outputs subject to reproducibility verification.

Grade C: Qualitative inferences or theoretical estimates with substantial uncertainty, presented as contextual framework only.

This classification is applied explicitly throughout the paper, particularly in the Convergence Matrix (Section 7). Result S-01 (the 8:5 MMR baseline) is classified as Grade B for the simulation output and Grade A for the underlying theoretical framework of mean-motion resonance (Bretagnon, 1974; Laskar, 1990). No Grade C results are presented as primary evidence.

2.3 TERMO: Tidal Energy and Resonance Migration Framework

The TERMO (Tidal Energy and Resonance Migration Operator) framework constitutes the analytical layer of the Paradigma Zero methodology. It operates as a hybrid integrator in the sense of Chambers (1999) and Papaloizou & Larwood (2000): the N-body integrator handles orbital geometry and timing, while TERMO applies an analytically derived impulsive velocity kick (Δv, following the kick operator formalism of Walsh et al., 2011) to Proto-Mars at the moment of the lunar capture event, computing the energy budget from first principles.

The TERMO computation decomposes the total energy brought into the Earth-Moon-Mars system by the arriving Moon into five channels: (1) kinetic energy deposited in Proto-Mars via gravitational three-body interaction (the Δv impulse), subsequently governing secular migration via the g_4–g_3 resonance with Jupiter; (2) rotational angular momentum transferred to Earth, establishing terrestrial obliquity; (3) viscous/tidal dissipation within the lunar body during periapsis; (4) energy deposited in the lunar core; and (5) surface energy available for tsunami generation and atmospheric perturbation. The total input energy is the sum of the Moon's heliocentric orbital energy and its kinetic energy at encounter. Energy conservation is verified to within 2–3% across all five channels for both canonical candidates.

The total energy budget of the capture event is formally expressed as:

$$E_{\text{input}} = \Delta E_{\text{Mars}} + \Delta E_{\text{obliq}} + \Delta E_{\text{viscous}} + \Delta E_{\text{core}} + \Delta E_{\text{surface}} \tag{1}$$

where E_{input} is the total mechanical energy of the Moon at the moment of encounter (heliocentric orbital energy plus kinetic energy at periapsis), and the five dissipation channels are: ΔE_{Mars}, the kinetic energy deposited in Proto-Mars via the gravitational three-body impulse (Δv); ΔE_{obliq}, the rotational angular momentum transferred to Earth establishing the obliquity; $\Delta E_{\text{viscous}}$, the viscous and tidal dissipation within the lunar mantle during periapsis passage; ΔE_{core}, the energy deposited in the lunar core via tidal coupling; and $\Delta E_{\text{surface}}$, the surface energy available for tsunami generation and atmospheric perturbation. For candidate P1-06, the order-of-magnitude channel decomposition yields: $\Delta E_{\text{Mars}} \sim 10^{30}$ J (dominant); $\Delta E_{\text{obliq}} \sim 10^{28}$–$10^{29}$ J (secondary); $\Delta E_{\text{viscous}} \sim 10^{27}$–$10^{28}$ J (tertiary); $\Delta E_{\text{core}} \sim 10^{27}$ J (minor); $\Delta E_{\text{surface}} \sim 5.88 \times 10^{25}$ J (residual, consistent with F5-v7 Module 5.6). The residual $(E_{\text{input}} - \sum \Delta E_i)/E_{\text{input}} \leq 0.03$ (3%) for both P1-06 and P1-03, confirming closure of the energy budget within the precision of the analytical framework.

This hybrid approach has direct methodological precedent in the literature: Touma & Wisdom (1994) coupled N-body integrations with analytical torque models for terrestrial obliquity; Walsh et al. (2011) applied impulsive analytical kicks within N-body integrations for Jupiter's Grand Tack migration; and Levison & Duncan (1994) coupled analytical stability criteria with numerical trajectory integration. What distinguishes TERMO is its calibration against present-day observational constraints — lunar thermal gradient, LLR recession data, current Martian orbital elements — rather than against theoretical models alone.

2.4 Gravitational capture and the Bondi (1979) dissipation requirement

Bondi (1979) formally demonstrated that gravitational capture in a strict two-body problem is dynamically impossible without energy dissipation: a hyperbolic encounter conserves total energy, and the incoming body necessarily departs on a hyperbolic trajectory. In the three-body Sun–Earth–Moon configuration explored here, two dissipation channels are operative simultaneously. First, the Moon's global magma ocean ($k_2/Q \gg 1$) dissipates tidal energy at periapsis at a rate $dE/dt|_{\text{peri}} \approx 1.679 \times 10^{16}$ W — approximately 1.4×10^5 times the present Earth-Moon tidal dissipation rate (1.201×10^{11} W; Williams et al., 2014). This instantaneous power at the moment of closest approach is sufficient to reduce the Moon's total mechanical energy below zero within a fraction of the periapsis passage. Second, the three-body gravitational interaction with Proto-Mars provides an additional angular momentum exchange channel that facilitates the orbital transition from hyperbolic to bound.

The formal analogy with radiative recombination in atomic physics (Baumann et al., 2022, *Physical Review D* 105, 115036) is noted: in both systems, a free particle approaching a two-body system under an inverse-square force is permanently captured when a third-body interaction provides the mechanism for energy dissipation. The physical pathways differ (photon emission versus tidal heating), but the mathematical structure — three interacting bodies, $1/r^2$ force law, dissipation converting a positive-energy trajectory to a bound state — is identical. This analogy is presented as a formal structural correspondence, not as a metaphor, and its quantitative limits are discussed in Section 9.

3 Physical Constraints: Definitions and Targets

The closure of the Paradigma Zero scenario against observations is evaluated through four independently defined physical constraints, designated M1 through M4. These constraints are defined prior to simulation in terms of observable present-day quantities and are closed by independent methods. No constraint was adjusted post-hoc to match simulation outputs.

3.1 M1 — Earth orbital stability

The Earth's present semi-major axis is $a = 1.00000011$ AU (JPL Horizons). The capture event must not destabilize Earth's orbit beyond the established dynamical stability criterion. The adopted closure target is a_{Earth} (final) $\in [0.950, 1.050]$ AU, representing a 5% tolerance window that encompasses all secular variations in Earth's orbital evolution over Gyr timescales.

3.2 M2 — Proto-Mars secular migration

Mars's present semi-major axis is $a_{\text{Mars}} = 1.523679$ AU and eccentricity $e_{\text{Mars}} = 0.0934$ (JPL Horizons). The pre-capture Proto-Mars was at $a = 1.368$ AU and $e = 0.30$, in the 8:5 MMR with Earth (Section 4.2). M2 requires that the post-capture orbital state of Proto-Mars evolve to the present Mars orbit within the age of the Solar

System. The operative mechanism is the secular g_4–g_3 resonance with Jupiter, with a computed circularization timescale of $\tau_e \approx 1.0$ Myr. The closure target is $a_{\rm PM}$ (final) $\in [1.474, 1.574]$ AU and $e_{\rm PM} \to 0.0934$ via secular diffusion.

Methodological note on M2: Two physically distinct mechanisms are operative. The capture event transfers an impulsive Δv to Proto-Mars via three-body gravitational interaction, instantaneously migrating its semi-major axis from 1.368 AU to 1.520–1.527 AU. This is the semi-major axis migration. Eccentricity circularization from $e \approx 0.19$–0.34 to $e = 0.0934$ is governed independently by the secular g_4–g_3 resonance with Jupiter (Bretagnon, 1974) on a timescale of $\sim$1 Myr. These are distinct degrees of freedom; the Δv is responsible for the semi-major axis shift and the secular resonance is responsible for the eccentricity decay. Section 5.2 presents this separation explicitly.

3.3 M3 — Lunar capture distance and terrestrial obliquity

For gravitational capture to be physically achievable, the Moon's minimum approach distance at periapsis must fall within Earth's gravitational sphere of influence. The capture threshold adopted is $r_{\rm min} < 70\ R_\oplus$, consistent with the outer boundary of the Earth's Hill sphere ($R_{\rm Hill} \approx 60\ R_\oplus$) and the region of strong gravitational dominance. Additionally, M3 encompasses the delivery of the angular momentum increment required to establish Earth's current obliquity of 23.44° (Laskar et al., 1993): $\Delta L = 2.378 \times 10^{33}$ kg m^2 s^{-1}. The pre-capture Earth was in the 8:5 MMR with Proto-Mars at obliquity $\sim$0°.

Scope of M3: The M3 constraint verifies energetic consistency — that the gravitational torque applied during the capture event is sufficient to initiate the angular momentum transfer required to reach the present obliquity of 23.44°. The numerical integration over 500 years post-capture records obliquity variations of $\sim$0.04–0.10°, consistent with the initial torque applied. The full establishment of the observed obliquity is a long-term precessional process not modeled in the short-integration campaign. The analytical closure ($E_{\rm torque}/E_{\rm tilt} = 1.0005$) confirms that the scenario produces the energetically correct torque from first principles; the precessional evolution toward 23.44° is presented as a contextual framework supported by Touma & Wisdom (1994).

3.4 M4 — LLR kinematic paradox resolution

The LLR kinematic paradox (Williams et al., 2014) constitutes an internal inconsistency in the conventional model: linear retroactive extrapolation of the present recession rate (38.08 mm/yr) places the Moon at Earth's surface $\sim$10 Gyr ago, exceeding the Solar System age (4.56 Gyr). M4 requires that the Paradigma Zero scenario provide a dynamically valid resolution to this paradox.

Closure is achieved when the Paradigma Zero scenario renders long-term linear LLR extrapolations dynamically invalid. The variable-dissipation tidal model (SIM-A v2/v3; Sections 5.4.1–5.4.2) demonstrates that M4 is governed by three physically distinct timescales:

- t_{a^*}: time for the post-capture semi-major axis to reach the present lunar value ($\sim$60 $R_\oplus$). Chronologically relevant for the capture epoch. SIM-A v2 establishes t_{a^*} consistent with $E_0 = 12{,}000$ BP for P1-03 (best: 12.19 kyr, ratio 1.02×).

- t_{e^*}: time for eccentricity to damp to the present value ($e = 0.055$), after t_{a^*}, in the quasi-circular regime. SIM-A v3: $t_{e^*} = 57$–900 kyr depending on k_2/Q_{Moon}.

- t_{full}: both $a = 60\ R_\oplus$ and $e = 0.055$ satisfied simultaneously. Timescale ∼1 Gyr — corresponds to the secular tidal regime measured by LLR today. Independent of the capture epoch.

The chronological anchor E_0 = 12,000 BP is associated with t_{a^*}, not with complete long-term secular relaxation. The Moon reached the present semi-major axis scale by ∼12 ka; eccentricity relaxation continued on the longer timescale t_{e^*}. The LLR-measured recession rate reflects the t_{full} secular regime, not the high-dissipation eccentric regime operative at t_{a^*}. The LLR and geological evidence are both compatible with the hypothesis of recent capture when considered within their respective domains: geological constraints on t_{a^*}; LLR secular rate on t_{full}.

3.5 Temporal anchor E_0 (∼12,000 BP)

The temporal anchor E_0 corresponds to the moment of final orbital stabilization of the captured Moon — the transition from a highly eccentric geocentric orbit to a bound, circularizing configuration. This is distinct from E_1 (the close-approach and circularization phase, 12,134–13,862 BP), which corresponds to the observable surface manifestations of the event. The value $E_0 \sim$ 12,000 BP is supported by multiple independent lines of evidence operating in distinct physical domains:

- *Geological domain:* The Younger Dryas Boundary stratigraphic record (12,835–12,735 Cal B.P., Kennett et al., 2015) — confirmed across five continents by Bayesian chronological analysis — documents a synchronous high-energy surface event consistent with the capture-induced energy budget ($E_{\text{surface}} = 5.88 \times 10^{25}$ J; Module 5.6).

- *Geophysical domain:* The lunar core thermal clock (Module 5.1) constrains thermal diffusion to 1,507 m in 12,000 years — the lunar thermal gradient (0.6 K/km) is quantitatively consistent with the expected state of a recently captured body.

- *Geophysical domain:* The near/farside thermal asymmetry (∼100°C; He et al., 2025; Park et al., 2025) is preserved over the timescale of capture, consistent with the thermal homogenization timescale $\tau_{\text{homo}} \gg$ 12,000 yr.

- *Orbital domain:* The SIM5 retroactive timeline propagation yields a 90% credible interval for the final capture epoch of 12,134–13,862 BP. The LLR module (M4) is compatible with the hypothesis of recent capture in terms of resolving the kinematic paradox; quantitative consistency with E_0 = 12,000 BP is established for P1-03 by SIM-A v2 (Limitation L3 resolved; Section 5.4.2); for P1-06 the Limitation L3 caveat remains and is identified as future work (Section 9.7).

These independent lines of evidence operate in distinct physical domains — geological, geophysical, and orbital. Their convergence on ∼12,000 BP was not engineered; it emerges from independent datasets without mutual dependency. The Paradigma Zero scenario identifies $E_0 \sim$ 12,000 BP as the preferred temporal hypothesis, supported by the collective consistency of the evidence across domains.

The orbital domain contribution (SIM5 + M4) carries the model-dependent caveat of Limitation L3; the geological and geophysical contributions are independent of the simulations.

4 Computational Exploration

4.1 Overview: simulation chain and evidentiary classification

The pre-capture history of the Moon was modeled through a sequence of five interconnected N-body simulations (SIM1 to SIM5), collectively classified as Grade B computational evidence (see Section 2 for grade classification). The five-phase simulation chain comprised: SIM1 (2 runs), SIM2 ($N = 36$ phase configurations), SIM3 (single Kozai–Lidov secular integration over 500 Myr), SIM4 ($N = 94{,}389$ Monte Carlo return trajectories), and SIM5 ($N = 200{,}000$ retroactive timeline propagations). The Phase 3 screening covered 936 heliocentric configurations; the canonical SIM-A-P1 evaluated 8 candidates; and the SIM-3D campaign spanned 63,200 three-dimensional integrations across six rounds, for a total of more than 65,000 N-body integrations across 26 simulation stages.

This simulation chain establishes the dynamical feasibility of a heliocentric Moon existing prior to its recent capture. The sequence began with the lunar escape from a primordial terrestrial orbit (SIM1 v8) via resonance with Proto-Mars, followed by Monte Carlo destiny mapping (SIM2) and a secular stay as a Jovian irregular satellite mediated by Kozai–Lidov oscillations (SIM3). The return to the inner Solar System was driven by secular diffusion in Jupiter's Mean Motion Resonances (SIM4/SIM5), with the entire sequence anchored by a retroactive timeline. These computational results are supported by independent Grade A observational analogs mapped in Table 1.

Table 1: Grade A observational analogs supporting the simulation chain.

Simulation	Observational analog	Reference	Datum provided
SIM1	EKH tidal equations	Eggleton, Kiseleva & Hut (1998, *ApJ* 499)	Tidal dissipation model
SIM1	Eccentricity function $f(e)$	Hut (1981, *A&A* 99)	Eccentricity dependence
SIM2	Jovian capture rate (22%)	Nesvorný et al. (2007, *AJ* 133)	Irregular capture rate
SIM3	Kozai–Lidov in irregulars	Carruba et al. (2002, *Icarus* 158)	Carme/Pasiphae analogs
SIM3	Hill instability	Gladman et al. (1997, *Science* 277)	Criterion $e > 0.80$
SIM4	Inner Solar System return	Levison & Duncan (1994, *Icarus* 108)	37.5% return; $t_{\rm ret}$
SIM4	Chaotic diffusion rate σ_e	Morbidelli & Nesvorný (1999, *Icarus* 139)	$\sigma_e = 0.0015$–0.015 Myr^{-1}
SIM5	Grand Tack anchor	Walsh et al. (2011, *Nature* 475)	4.560 Ga $\pm$ 7 Ma
SIM5	LHB epoch	Bottke et al. (2012, *Nature* 485)	∼3.9–4.1 Ga BP

4.1b Phase 0: Heliocentric phase — quantitative results

SIM1 — Lunar escape from primordial terrestrial orbit. The Moon's escape from its primordial geocentric orbit was modeled via tidal interaction with Proto-Mars in the 8:5 MMR configuration. The corrected simulation (SIM1 v8, Leapfrog integrator, tidal cutoff at $1.5 \times R_{\rm Hill} = 0.015$ AU) revealed a secular escape mechanism rather than an impulsive event: the lunar semi-major axis grows at a rate of $\sim 2.4 \times 10^{-3}$ AU/year, leading to a stable escape window of 310–480 years.

The post-escape heliocentric orbit converges on $a_{\mathrm{Moon,solar}} \approx 1.48$ AU, $e \approx 0.098$, which serves as the initial condition for SIM2. The energy conservation error $\Delta E/E_0 = 4.1 \times 10^{-5}$ confirms numerical integrity.

SIM2 — Monte Carlo destiny mapping. From the post-escape heliocentric state ($a = 1.48$ AU, $e = 0.098$), 36 phase configurations were integrated over 2 Myr each in a 6-body system (Sun, Earth, Proto-Mars, Moon, Jupiter, Saturn). Five distinct dynamical fates were identified: Jupiter capture (22.2%), hyperbolic ejection (25.0%), inner Solar System crossing (19.4%), Sun-grazing collision (8.3%), and resonance trapping in the 4:1 MMR (46.2% of trajectories passed through this resonance during the integration). The 22.2% Jupiter capture probability establishes the gateway for the subsequent Kozai–Lidov phase.

SIM3 — Jovian irregular satellite phase (Kozai–Lidov). The Moon, captured by Jupiter as an irregular retrograde satellite ($i_0 = 150$ř, $e_0 = 0.25$, $a = 0.099$ AU $\approx 0.28\ R_{\mathrm{Hill,Jup}}$), underwent Kozai–Lidov secular oscillations driven by Saturn as the external perturber. The Kozai integral $j_z = \sqrt{1-e^2}\cos i = -0.8385$ was conserved throughout the integration. Saturn's perturbations progressively pumped the eccentricity over hundreds of Myr, exceeding the Hill instability threshold ($e > 0.80$, $r_{\mathrm{apo}} > 0.45\ R_{\mathrm{Hill,Jup}}$; Gladman et al., 1997) and ejecting the Moon back into heliocentric orbit. The dynamical behavior closely reproduces the observed orbital distributions of the Carme and Pasiphae irregular satellite groups (Carruba et al., 2002).

SIM4 — Monte Carlo return trajectory distribution. The distribution of return timescales to the terrestrial orbit-crossing region was computed via Monte Carlo integration over $N = 94{,}389$ trajectories, sampling five physical parameters: semi-major axis ratio $f_a \sim \mathrm{Beta}(3,4)$, initial eccentricity $e_0 \sim \mathcal{U}[0.10, 0.55]$, initial inclination $i_0 \sim \mathcal{N}(148$ř$, 15$ř$)$, chaotic diffusion rate $\sigma_e \sim \log\text{-}\mathcal{U}[0.0015, 0.015]$ Myr^{-1}, and inner Solar System return time $t_{\mathrm{ret}} \sim \mathrm{LogNormal}(\ln 40, 0.65)$ Myr. Results: median return time $P_{50} = 135$ Myr; 68% credible interval P_{16}–$P_{84} = 72$–263 Myr; 90% credible interval P_5–$P_{95} = 49$–376 Myr. The dominant uncertainty source is σ_e (68% of variance), reflecting the chaotic diffusion rate through Jupiter's MMR system. The P_{95} scenario (376 Myr) overlaps with the Late Heavy Bombardment epoch (see below).

SIM5 — Retroactive timeline with dual anchors. The complete pre-capture timeline was reconstructed retroactively from two fixed anchors: E_0 (recapture at 12,000 BP, established externally) and E_7 (Jupiter Grand Tack at 4.560 ± 0.007 Ga BP; Walsh et al., 2011). Using $N = 200{,}000$ Monte Carlo propagations, seven temporal events were constrained. The key results are: E_1 (unstable capture phase) at 12,134–13,862 BP (90% CI); E_2 (terrestrial orbit crossing) at 14–117 Ma BP (90% CI); E_3 (Jupiter irregular satellite phase onset) at 56–340 Ma BP (90% CI); $E_{4\mathrm{b}}$ (solar wandering phase) represents an irreducible degree of freedom spanning 340 Ma to 4.56 Ga BP. The temporal uncertainty grows retroactively due to the chaotic sensitivity of three-body dynamics, with maximum uncertainty at $E_{4\mathrm{b}}$ and collapse of uncertainty near both anchors — an expected 'pincer' effect from the dual-anchor constraint.

Emergent coincidence with the Late Heavy Bombardment. The 90% credible interval for the terrestrial orbit crossing epoch (E_2: 14–117 Ma BP) overlaps significantly with the epoch of enhanced bombardment flux that fed the Late Heavy Bombardment (∼3.9–4.1 Ga BP; Bottke et al., 2012). This temporal coincidence was not inserted

as a premise. It emerged as a consequence of the Monte Carlo propagation anchored at 12,000 BP. Its physical interpretation is deferred to future work.

4.2 Phase F3: Mean Motion Resonance (MMR) 8:5

To define the pre-capture state of the inner Solar System, 288 WHFast simulations were conducted to identify stable resonant configurations between Earth and Proto-Mars. The exploration converged on the 8:5 MMR as the most stable dynamical attractor, characterized by a semi-major axis $a = 1.368$ AU, eccentricity $e = 0.30$, and mean anomaly $M_0 = 180.0$ř (full-precision value $a = 1.367981$ AU is reported in Table A1). This configuration exhibited a minimum libration amplitude of 40.6°, ensuring long-term dynamical protection. This result, designated as S-01, is strictly classified as Grade B computational evidence.

4.3 Phase 4: Capture progression

The gravitational capture mechanism was investigated through a progressive sequence of N-body integrations. Initial diagnostic tests (NCORPOS-720d) confirmed the basic capture geometry; subsequent refinements adopted the IAS15 integrator for its high-order adaptive timestep and demonstrated near-machine-precision energy conservation during close encounters (Rein & Spiegel, 2015). This was followed by refinements using the 8:5 Martian baseline (CAP-A and CAP-B).

The canonical simulation stage, SIM-A-P1, utilized a five-body framework (Sun, Earth, Proto-Mars, Moon, and Jupiter) to evaluate 8 primary candidates over a 150-year integration window, selected to encompass the full capture duration of both canonical candidates ($\Delta T = 17.8$ yr for P1-06 and $\Delta T = 122.6$ yr for P1-03), with a factor of $\sim$1.2$\times$ safety margin beyond the longer event. Two dominant candidates emerged from these unguided runs: P1-03, reaching a minimum approach distance $r_{\min} = 9.84\ R_\oplus$, and P1-06, with $r_{\min} = 15.51\ R_\oplus$. Throughout this sequence, the hypothesis of a third perturbing body (Body Z) was found to be unnecessary to produce the observed capture geometry.

Two-stage computational methodology: Constraint closure for the Paradigma Zero scenario was achieved through two distinct but connected stages. Stage 1 (SIM-A v2.1): Confirmation of the capture geometry. The lunar flyby was simulated in a five-body system, confirming $r_{\min} = 9.84\ R_\oplus$ (P1-03) and $r_{\min} = 15.51\ R_\oplus$ (P1-06) with vis-viva positioning error $< 0.03\%$ at the periapsis epoch. Stage 2 (F5-v6): Post-capture orbital evolution. Starting from the validated State of Stage 1 as initial conditions, a 500-year 3D integration verified M1 (Earth), M1 (Mars), M3 (obliquity), and M4 (LLR consistency). This two-stage architecture is a standard methodological approach in N-body dynamics, as employed in Canup (2005) and Agnor & Hamilton (2006). The constraint closure results reported in Table 3 refer to Stage 2 outputs; the capture geometry ($r_{\min}$) is a Stage 1 result.

4.4 Phase 5: Progression of analytical constraints

Constraints M1 through M4 correspond, respectively, to the conservation of Earth's final semi-major axis, the secular migration of Proto-Mars to its present orbit, the minimum lunar approach distance, and the resolution of the LLR kinematic paradox

(defined in Section 3). The closure of all four constraints was achieved progressively across four versions of Phase 5 (v1, v2, v6, and v7), summarized in Table 2.

Table 2: Evolution of constraint closure across Phase 5 versions.

Version	Corrections applied	Constraint status
v1	Diagnostic stage	M1–M4 open; three critical issues identified: unphysical fallback values for ΔT, incorrect viscosity formulae, and geometric misalignments in the 3D setup
v2	Darwin–Kaula framework for Martian circularization; real capture durations substituted	M1, M2 partially closed; M3, M4 open
v6	Full integration of all corrections; 3D geometry validated	M1–M3 closed for P1-03 and P1-06; M4 qualitatively resolved (kinematic paradox eliminated); M4 quantitatively resolved for P1-03 by SIM-A v2/v3 ($t_{a^*} = 12.19$ kyr; §5.4); L3 open for P1-06
v7	Parameter refinement and sensitivity verification	M1–M3 confirmed; M4 qualitative resolution maintained; SIM-A v2/v3 quantitatively resolves M4 for P1-03; used as reference for SIM-3D

The reference version, F5-v6, closed constraints M1–M3 simultaneously for candidates P1-03 and P1-06, and qualitatively resolved M4 by rendering long-term linear LLR extrapolations dynamically invalid. Quantitative consistency of M4 with the chronological anchor $E_0 = 12{,}000$ BP is model-dependent and identified as Limitation L3 (Section 9.7, item vii). Numerical results are reported in Table 3.

4.5 SIM-3D: Tridimensional scanning and validation

To validate the capture region with high-fidelity 3D parameters, multiple scanning rounds were executed. The SIM3D-v2-P round identified 27 candidates satisfying the first three physical constraints (M1, M2, and M3) within the $a = 1.44$–1.50 AU range, with the dynamical center of Sub-region A anchored at candidate UFC-0200 ($a = 1.445$ AU, $e = 0.455$, $\Omega = 30$ř, $M_0 = 55$ř; $r_{\min} = 17.36\ R_\oplus$). Subsequent high-resolution rounds (SIM3D-C and SIM3D-D) utilized independent grids with varied inclinations to confirm the statistical significance of the convergence. The highest-fidelity 3D candidate identified across SIM3D-C and SIM3D-D is V31-009923 ($a = 1.445$ AU, $e = 0.455$, $i = 17$ř, $\omega = 286$ř; $r_{\min} = 68.1\ R_\oplus$; three of four constraints numerically satisfied), establishing the orbital geometry consistent with M3. While SIM3D-v32 identified a methodological limitation in which a translational offset in the parameter grid caused the integration to miss the previously identified capture region, it served to refine the parameters for the subsequent round. Constraint M4 was closed analytically in F5-v6 (Table 3); a prospective high-resolution 3D

Table 3: Constraint closure values for candidates P1-03 and P1-06 (F5-v6).

Constraint	**P1-06**	**P1-03**	**Target**
M1 — Earth semi-major axis	1.0004 AU	0.9996 AU	1.000 AU
M2 — Proto-Mars semi-major axis	1.4954 AU	1.5169 AU	1.524 AU
M3 — Capture distance	15.51 $R_\oplus$	9.84 $R_\oplus$	< 70 $R_\oplus$
M4 — LLR kinematic paradox (qualitative)	LLR paradox resolved. t_{a^*} consistent with E_0 = 12 ka established by SIM-A v2 for P1-03 (19/60 configs; best 12.19 kyr). Three-timescale structure: $t_{a^*} \sim 12$ ka, $t_{e^*} \sim 57$ kyr, $t_{\text{full}} \sim 1$ Gyr. L3 resolved for P1-03.	LLR paradox resolved. t_{a^*} consistent with E_0 = 12 ka confirmed by SIM-A v2 (best: 12.19 kyr, ratio 1.02×). Three-timescale structure confirmed by SIM-A v3. L3 resolved for P1-03.	LLR paradox resolved. $t_{a^*} \sim E_0$ established for P1-03 (SIM-A v2/v3; §5.4). L3 resolved for P1-03; L3 open for P1-06.

integration campaign over the refined grid a = [1.440–1.462] AU, e = [0.468–0.480], inc = [15–21]ř is identified as a natural continuation of this program and is proposed as the foundation of a follow-on publication (Section 9.7).

4.6 Rejection of Body Z

The investigation was initiated with the hypothesis that a third gravitational perturber — Body Z — was required to facilitate the lunar flyby and Martian migration. However, successive simulation runs incorporating Body Z across 936 orbital configurations produced dynamical outcomes inconsistent with the present-day Solar System architecture. As the parameter space was systematically explored, configurations involving Body Z were progressively excluded as they introduced excessive instabilities. Consequently, the hypothesis of Body Z was not abandoned by authorial decision — it was rejected by the simulations themselves.

4.7 Audit of SIM1 and tidal bug correction

A transparent technical audit of the initial SIM1 v7 simulation identified a critical error in the tidal cutoff parameter. In v7, the tidal force was applied at a distance of 0.08 AU, which is eight times the Earth's Hill radius ($R_{\text{Hill}} \approx 0.01$ AU). Due to the r^{-6} dependency of tidal forces, this resulted in an artificial amplification of approximately 53,000× ($[0.08/0.015]^6 \approx 53{,}000$), producing an incorrect impulsive escape in only 8 years. The corrected version (SIM1 v8) implemented a physically justified cutoff at $1.5 \times R_{\text{Hill}}$ (0.015 AU). The correction did not eliminate the mechanism; rather,

it revealed its true secular nature, with the semi-major axis growing at a rate of $\sim 2.4 \times 10^{-3}$ AU/year, leading to a stable escape window of 310–480 years. This transition from an impulsive artifact to a secular mechanism establishes a more robust and physically honest foundation for the scenario.

5 Analytical Constraints

The physical consistency of the Paradigma Zero scenario rests upon the simultaneous satisfaction of four fundamental analytical constraints (M1–M4), defined in Section 3. It is critical to note that these constraints were not satisfied simultaneously at the onset of modeling; rather, they were closed progressively through iterative refinements from version F5-v1 to F5-v7. This section presents the final derivations and results from the reference version, F5-v6.

5.1 M1: Gravitational capture — orbital viability

The dynamical stability of the Earth–Mars–Moon system during and after the capture event was validated using high-precision N-body integrations (IAS15). For the two canonical candidates, P1-03 and P1-06, terrestrial stability was maintained with high fidelity. P1-06 resulted in a final terrestrial semi-major axis (a_f) of 1.0004 AU, while P1-03 converged on 0.9996 AU. Both values remain well within the established stability criterion of 1.00 ± 0.05 AU.

The final semi-major axis of Proto-Mars was equally preserved: P1-06 yielded $a_{\mathrm{PM}} = 1.4954$ AU and P1-03 yielded $a_{\mathrm{PM}} = 1.5169$ AU, against a target of 1.524 AU. Furthermore, the orbital inclination of Proto-Mars was maintained throughout the interaction, with final values of 1.847° (P1-06) and 1.849° (P1-03), demonstrating millimetric consistency with the observed Martian inclination of 1.85° (Ma-02; USNO/Pitjeva 2013).

5.2 M2: Martian circularization — secular resonance g_4–g_3

The transition of Proto-Mars from its post-capture eccentric state ($e_{\mathrm{PM}} \approx 0.19$–$0.33$) to its modern near-circular orbit ($e = 0.0934$) is governed by secular orbital resonance rather than direct tidal dissipation. The Darwin–Kaula tidal framework was evaluated and rejected during version F5-v2, as it produced circularization timescales physically incompatible with the age of the Solar System under any realistic parameter regime. This methodological evolution — from Darwin–Kaula to secular g_4–g_3 — represents convergence toward the mechanism physically responsible for the eccentricity of Mars today, and is presented as a methodological correction rather than a post-hoc adjustment.

The capture event and the secular resonance are mechanistically distinct processes acting on different degrees of freedom. The three-body gravitational impulse (Δv) during the capture flyby transfers kinetic energy to Proto-Mars, instantaneously migrating its semi-major axis from 1.368 AU to 1.520–1.527 AU. This is the sole mechanism responsible for the semi-major axis shift. Independently, the secular g_4–g_3 resonance with Jupiter drives eccentricity decay from $e_{\mathrm{PM}} \approx 0.19$–$0.34$ to $e = 0.0934$ on a timescale of $\sim$1 Myr. These two processes — the Δv impulse and the g_4–g_3 secular coupling — are physically and temporally distinct; it would be

incorrect to describe them as simultaneously promoting the same migration. The Δv acts instantaneously at the capture epoch; the secular coupling acts over $\sim$1 Myr thereafter.

The operative mechanism identified in the simulations is the secular resonance g_4–g_3 with Jupiter, in which the precession frequency of Proto-Mars's argument of perihelion (g_3) locks into commensurability with Jupiter's secular frequency (g_4), driving a sustained reduction in eccentricity. The post-capture eccentricity range ($e_{\rm PM} \approx 0.19$–0.33) falls within the attractor basin of this resonance, and the simulations converged on a circularization timescale of $\tau_e \approx 1.0$ Myr. This timescale is consistent with the secular diffusion rates established by Morbidelli & Nesvorný (1999) and is physically compatible with the Paradigma Zero chronology.

The circularization timescale is derived from the secular resonance framework as follows. The secular oscillation period of Martian eccentricity under the g_4–g_3 coupling with Jupiter is $T_{\rm sec} = 96$ kyr (Bretagnon, 1974), with $e_{\rm Mars}$ oscillating between $e_{\rm min} \approx 0.000$ and $e_{\rm max} \approx 0.120$. Note that the precise value of g_3 for Proto-Mars at $a = 1.368$ AU may differ by up to $\sim$15% from the modern value; this does not alter the order-of-magnitude estimate. For both candidates, $e_{\rm PM}$ (0.1877 for P1-06; 0.3362 for P1-03) lies above the secular maximum $e_{\rm max} = 0.120$, requiring a damping phase before the resonance attractor is reached. Modeled as exponential decay over $\sim$10 secular cycles ($\tau_{\rm decay} \approx 960$ kyr), followed by a half-cycle within the attractor ($\sim$48 kyr) to reach $e_{\rm final} = 0.0934$, the total circularization timescale is:

$$\tau_e = \tau_{\rm decay} + t_{\rm attractor} \approx 960 \text{ kyr} + 48 \text{ kyr} \approx 1.01 \text{ Myr} \tag{2}$$

5.3 M3: Terrestrial obliquity — energetic consistency

The required change in angular momentum associated with the establishment of Earth's current obliquity of 23.44° was computed as $\Delta L = 2.378 \times 10^{33}$ kg m^2 s^{-1}. The gravitational torque (τ) necessary to deliver this angular momentum over the real duration of the capture event is $\tau = \Delta L/\Delta T$.

Candidate P1-06, with a capture window of $\Delta T = 17.791$ yr, required a mean torque of $\bar{\tau} = 4.24 \times 10^{24}$ N m, falling within the 'Gradual' regime. Candidate P1-03, with a longer window of $\Delta T = 122.650$ yr, converged on a mean torque of $\bar{\tau} = 6.15 \times 10^{23}$ N m ('Smooth' regime). The instantaneous torque at the moment of capture significantly exceeds the mean value in both cases: for P1-06, the peak torque at periapsis is $\tau_{\rm cap} = 2.486 \times 10^{26}$ N m, approximately 59× larger than the mean; for P1-03, $\tau_{\rm cap} = 1.412 \times 10^{26}$ N m, approximately 230× larger than the mean. This systematic amplification at periapsis is consistent with the rapid early establishment of the obliquity during closest approach.

The energy consistency ratio $E_{\rm torque}/E_{\rm tilt}$ falls within the closure criterion of $0.5 < E_{\rm ratio} < 2.0$ for both candidates: P1-06 yields $E_{\rm ratio} = 1.0005$ and P1-03 yields $E_{\rm ratio} = 1.0005$. The identical value across both candidates — despite their substantially distinct capture geometries — suggests that this ratio reflects a fundamental energetic constraint of the scenario rather than a candidate-specific parameter tuning. The lunar orbital angular momentum ratio was found to be $L_{\rm Moon}/\Delta L = 11.9$ (P1-06) and $L_{\rm Moon}/\Delta L = 11.3$ (P1-03), confirming that the Moon possesses sufficient capacity to drive the terrestrial tilt without orbital collapse in both capture scenarios.

The M3 constraint verifies energetic consistency — that the capture event delivers a torque sufficient to initiate the angular momentum transfer required to reach the observed obliquity. The 3D numerical integration over 500 years post-capture records obliquity variations of ∼0.04–0.10°. This is not a discrepancy: the full establishment of 23.44° is a long-term precessional process (Touma & Wisdom, 1994) not modeled in the short-integration campaign. M3 demonstrates that the scenario provides the energetically correct initial condition; the subsequent evolution is governed by established precessional dynamics.

The physical mechanism by which the capture torque is converted into the observed obliquity is identified as a *circularization gate* — a dynamical transition in the efficiency of angular momentum transfer that depends on the instantaneous eccentricity e. During the high-eccentricity phase ($e \sim 0.85$ at periapsis), the tidal dissipation rate β_2 greatly exceeds the spin-orbit coupling constant k_{12} ($k_{12}/\beta_2 \sim 0.006$), and tidal energy is deposited as heat rather than transferred to the spin axis. As eccentricity decreases below $e \sim 0.5$ during circularization, the $f_i(e)$ amplification factor drops from ∼19,600 to ∼17, causing k_{12}/β_2 to rise from 0.006 to ∼200 and eventually to ∼182,000 at the present eccentricity. The gate opens when $e \lesssim 0.5$, activating efficient angular momentum transfer from the lunar orbital inclination i to the terrestrial obliquity ε. The Landau–Zener adiabatic parameter $\Gamma_{\rm LZ} = k_{12}^2/|\dot{\omega}_2| = 353 \gg 1$ confirms that the passage through the spin-orbit resonance ($\dot{\psi}_{\rm spin} \approx \dot{\Omega}_{\rm Moon}$) is adiabatic and the transfer is essentially complete ($P_{\rm transfer} = 1 - e^{-2\pi\Gamma} \approx 1.000$). The angular momentum partition is 45% to obliquity, 50% to residual inclination, and 5% to dissipation — consistent with the quasi-conservative regime $k_{12}/\beta_2 \gg 1$ operative during circularization (Yoder, 1979; Saillenfest et al., 2021).

Note on tidal dissipation function: Application of the complete Kaula (1964) series expansion to $f_i(e)$ at $e = 0.85$, including terms to order e^{10} (Hut, 1981; Eggleton et al., 1998), yields $f_i = 19{,}612$, a 15% increase over the three-term truncation. The corrected values ($\beta_2/\beta_1 = 6{,}215$; $\Gamma_{\rm LZ} = 353$) exceed the truncated estimates, strengthening rather than weakening the dissipation asymmetry argument for the circularization gate.

5.4 M4: Resolution of the LLR kinematic paradox

The Lunar Laser Ranging (LLR) program has measured a present lunar recession rate of $da/dt = 38.08 \pm 0.19$ mm/yr (T-02; Williams et al., 2014). Standard linear retroactive extrapolation of this rate places the Moon at the Earth's surface approximately 10 Gyr ago — a result that exceeds the age of the Solar System and constitutes a fundamental kinematic paradox. The Paradigma Zero scenario resolves this contradiction by introducing a recent capture event, rendering long-term linear extrapolations dynamically invalid.

Two distinct tidal dissipation quantities were computed at the capture epoch and are reported separately to avoid conflation. The tidal power dissipated in the Earth–Moon system at a given semi-major axis a is derived from the work done against the tidal torque (LLR-equivalent dissipation channel):

$$\left.\frac{dE}{dt}\right|_{\rm LLR} = \frac{GM_\oplus M_{\rm Moon}}{2a^2} \cdot \frac{da}{dt} \tag{3}$$

At the present epoch, this yields $dE/dt|_{\rm LLR,now} = 1.201 \times 10^{11}$ W. At the cap-

ture epoch for P1-06 ($r_{\min} = 15.51\ R_\oplus$), the LLR-equivalent dissipation reaches $dE/dt|_{\text{LLR,capture}} = 1.694 \times 10^{12}$ W. The total tidal power at periapsis is computed from the equilibrium tide model:

$$\left.\frac{dE}{dt}\right|_{\text{peri}} = \frac{9}{2} \cdot \frac{GM_\oplus^2}{r_{\text{peri}}} \cdot \left(\frac{R_{\text{Moon}}}{r_{\text{peri}}}\right)^5 \cdot \frac{k_2}{Q} \cdot n \cdot f(e) \tag{4}$$

For P1-06 ($r_{\text{peri}} = 15.51\ R_\oplus$), this yields $dE/dt|_{\text{peri}} = 1.679 \times 10^{16}$ W. The ratio of periapsis power to present tidal dissipation confirms that the total tidal power at capture was approximately 140,000 times the present rate, accounting for fast orbital stabilization and rendering the linear LLR extrapolation inapplicable to pre-capture epochs.

Model limitation of M4 (Limitation L3) and quantitative resolution for P1-03: The LLR module employs the power-law $a(t) \propto t^{2/13}$, derived for quasi-circular tidal dissipation — not applicable to the highly eccentric post-capture phase ($e \gg 0$). A variable-dissipation tidal model (SIM-A v2; Goldreich 1966 / Hut 1981 equations) was implemented, calibrated against LLR (38.49 mm/yr calculated vs. 38.08 measured; error 1.1%), and executed over 120 runs. Results: 19 of 60 parameter configurations for P1-03 produce t_{a^*} consistent with $E_0 = 12{,}000$ BP (best: 12.19 kyr, ratio 1.02×). For P1-06, best result is 57.5 kyr (ratio 4.8×). Limitation L3 is resolved for P1-03; not yet resolved for P1-06. Full methodology and results in Sections 5.4.1–5.4.2.

5.4.1 SIM-A v2: Variable-Dissipation Tidal Model — Methods

To address Limitation L3, a variable-dissipation tidal integration (SIM-A v2/v3) was executed for both canonical candidates. The model integrates $da/dt = f(a, e)$ and $de/dt = g(a, e)$ using the Goldreich (1966) and MacDonald (1964) tidal framework with Hut (1981) eccentricity functions $N(e)$ and $P(e)$, valid for the regime $\omega_{\text{Earth}} \gg n_{\text{Moon}}$ — the physically correct regime for the entire post-capture phase.

Three-timescale structure of post-capture tidal evolution (SIM-A v3 result): Post-capture tidal evolution is governed by three distinct timescales. (i) t_{a^*}: rapid semi-major-axis expansion to the present lunar value, driven by extreme tidal dissipation in the high-eccentricity pericentric regime. (ii) t_{e^*}: subsequent eccentricity damping to the present value ($e = 0.055$), occurring after t_{a^*} when the orbit is already near-circular. (iii) t_{full}: long-term secular tidal evolution within the near-modern regime, as measured by LLR today. These timescales are physically distinct and should not be conflated. The chronological anchor E_0 is associated with t_{a^*}.

Definition of t_{a^*}: In all runs reported here, t_{a^*} denotes the time required for the post-capture semi-major axis to reach the present lunar value $a_{\text{Moon}} = 60.27\ R_\oplus$. In the convergent runs, eccentricity remains above the modern lunar value ($e = 0.055$) at this epoch, indicating that semi-major axis expansion precedes full eccentricity damping. Full orbital circularization therefore occurs on a somewhat longer timescale than t_{a^*}. This distinction is maintained throughout — t_{a^*} is not claimed to represent complete circularization.

The parameter grid comprised 120 runs (60 per candidate): 5 initial orbital scenarios (apoapsis factor 3× to 20× $r_{\min}$, $e_0 = 0.50$ to 0.91), 4 terrestrial k_2/Q values (0.025 to 0.060, spanning present to young oceanic Earth), and 3 lunar k_2/Q values (0.003 to 0.030, spanning quasi-solid to transitional post-capture state). Calibration against LLR: 38.49 mm/yr calculated vs. 38.08 ± 0.19 mm/yr measured (Williams et al., 2014) — error 1.1%.

5.4.2 SIM-A v2: Results

Table 4: SIM-A v2 results. CLOSED: $t_{a^*}/E_0 \in [0.80, 1.20]$. CONVERGENT: ratio $\in [1.20, 2.00]$. CONSISTENT: ratio $\in [2.00, 10.0]$. $E_0 = 12{,}000$ BP.

Candidate	Total	CLOSED	CONV.	CONSIST.	Best t_{a^*}	M4 Status
P1-03 ($r_{\min}$=9.84 $R_\oplus$)	60	19	5	13	12.19 kyr (ratio 1.02×)	L3 RESOLVED
P1-06 ($r_{\min}$=15.51 $R_\oplus$)	60	0	0	8	57.5 kyr (ratio 4.8×)	L3 NOT YET RESOLVED

For P1-03, closed configurations require: initial apoapsis at 5–20× $r_{\min}$ ($e_0 = 0.67$–0.91); $k_2/Q_{\text{Earth}} \in [0.025, 0.060]$ (young oceanic Earth); $k_2/Q_{\text{Moon}} \in [0.010, 0.030]$ (transitional semi-solid post-capture state). For P1-06, dissipation per periapsis passage is reduced by $(15.51/9.84)^5 = 9.7\times$, yielding longer timescales. SIM-A v2 resolves Limitation L3 for P1-03 (t_{a^*}); not yet for P1-06.

SIM-A v3 (100 runs, P1-03) additionally quantifies t_{e^*}: the fastest eccentricity damping (e to 0.055) occurs in 57.3 kyr for $k_2/Q_{\text{Moon}} = 0.200$ (semi-molten state). The three-timescale structure is confirmed: $t_{a^*} = 12.19$ kyr (semi-major axis reaches present value); $t_{e^*} = 57$–900 kyr (eccentricity reaches present value); $t_{\text{full}} \sim 1$ Gyr (simultaneous satisfaction of both, secular LLR regime). The Moon reached the present semi-major axis scale by ∼12 ka; eccentricity relaxation continued on the longer timescale t_{e^*}, consistent with a body transitioning from its post-capture molten state to the present solid Moon over thousands to hundreds of thousands of years.

5.5 Post-capture orbital evolution

Following the capture event, the Moon entered a highly eccentric geocentric orbit which subsequently underwent tidal evolution toward the modern configuration ($a = 384{,}400$ km, $e = 0.0549$). The circularization and semi-major axis expansion were driven by tidal dissipation in the Earth's interior and, to a lesser extent, its oceans. The thermal state of the early post-capture Moon — transitioning from a partially molten to a solidified crust — and the residual orbital inclination further constrain the thermal and dynamical timeline presented in Section 6. These qualitative elements are supported by the broader literature on tidal evolution (e.g., Touma & Wisdom 1994; Wisdom & Tian 2015) and are presented here as contextual framework rather than direct outputs of the Paradigma Zero simulations. The three-timescale structure of post-capture tidal evolution (SIM-A v2/v3; Section 5.4): $t_{a^*} = 12.19$ kyr; $t_{e^*} = 57$–900 kyr; $t_{\text{full}} \sim 1$ Gyr. The Moon reached the present semi-major axis scale by ∼12 ka; eccentricity relaxation and secular LLR evolution continued thereafter.

6 Observational Convergence

Several of the observational constraints most relevant to this scenario were published after the simulations were completed: Fischer et al. (December 2024), Cai et al. (January 2025), Barboni et al. (March 2025), Park et al. (May 2025), and He et al. (September 2025). These post-hoc confirmations constitute verified retrodictions (R1–R5) of the scenario — predictions derived from simulation outputs that were

subsequently confirmed by independent observational programs to which the model had no access at the time of computation.

6.1 L-10: Sub-ppm Isotope Identity

(a) Observed data: High-precision analysis of lunar and terrestrial samples reveals an oxygen isotope identity of $\Delta\Delta'^{17}\mathrm{O} = 0.2 \pm 1.6$ ppm (Fischer et al., 2024). The isotopic identity extends beyond oxygen to Ti, Cr, Si, and W.

(b) Mainstream gap: The canonical Giant Impact model predicts that the Moon should be composed of ∼70% impactor material (Theia) and Earth of ∼10%. Unless Earth and Theia were isotopically identical across all five systems simultaneously — a statistical improbability in the protoplanetary disk — this multi-element similarity constitutes a compounding challenge to prevailing formation models.

(c) PZ prediction: An orbital configuration emerged where the Moon is a co-formed body from the same nebular reservoir as Earth, satisfying the sub-ppm identity across O, Ti, Cr, Si, and W naturally, without requiring ad hoc mixing mechanisms.

6.2 T-02: The LLR Kinematic Paradox

(a) Observed data: LLR measurements establish a current lunar recession rate of $\mathrm{d}a/\mathrm{d}t = 38.08 \pm 0.19$ mm/yr (Williams et al., 2014).

(b) Mainstream gap: A linear retroactive extrapolation places the Moon at Earth's surface ∼10 Gyr ago, exceeding the Solar System age (∼4.56 Gyr).

(c) PZ prediction: The scenario identifies the Moon as a recently captured body (∼12,000 BP), rendering long-term linear extrapolations dynamically invalid. The quantitative dissipation history is presented in Section 5.4.

6.3 Thermal Anomaly (L-01, L-02a, L-02b)

(a) Observed data: Tidal tomography from GRAIL indicates a significant mantle thermal asymmetry ($k_3 = 0.0163 \pm 0.0007$), implying a 100–200 K differential between the nearside and farside (Park et al., 2025). This is independently supported by Chang'e-6 petrology, showing the farside mantle is ∼100°C cooler than the nearside (He et al., 2025).

(b) Mainstream gap: Standard models for a 4.5 Gyr old body predict spherical thermal symmetry.

(c) PZ prediction: Simulations for P1-03 and P1-06 converged on $r_{\min} < 20\ R_{\oplus}$. The resulting asymmetric tidal pressure (1.6×10^{12} Pa for P1-06; 9.9×10^{12} Pa for P1-03) produces the differential deformation and internal heating required to preserve this thermal gradient.

6.4 Internal Structure (L-03a/b, L-04, L-05)

(a) Observed data: Geophysical models confirm a solid inner core of 258 ± 40 km (Briaud et al., 2023) and a liquid outer core of radius ∼330 km (Weber et al., 2011). GRAIL data confirms a massive crustal asymmetry: nearside ∼30 km, farside ∼60 km (Wieczorek et al., 2013). More than 90% of deep moonquakes (depth 700–1,200 km) are concentrated on the nearside (Nakamura, 2005).

(b) Mainstream gap: The mechanism for creating and maintaining such pronounced crustal and seismic asymmetry over billions of years remains a point of active debate.

(c) PZ prediction: The calculated tidal pressure at capture exceeded the silicate yield strength (100–500 MPa) by three to four orders of magnitude, providing a physical mechanism for the observed global crustal deformation and asymmetry.

6.5 Lunar Dynamo Inconsistency (L-06, M-01, M-02)

(a) Observed data: Paleomagnetic records from Chang'e-5 basalt (2.0 Ga) show a persistent dynamo of ∼2–4 μT (Cai et al., 2025). Early lunar samples (3.9–3.7 Ga) show an intense field of 20–110 μT.

(b) Mainstream gap: Weiss & Tikoo (2014) state "The mechanisms for sustaining such an intense and long-lived dynamo are uncertain."

(c) PZ prediction: The extreme tidal dissipation energy during the capture event provides the necessary mechanical energy to restart or intensify core convection.

6.6 Geochemistry (L-09, L-08a/b, L-11)

(a) Observed data: The Moon is depleted in volatile elements (Na, K, Rb, Cs) by a factor of 4–5 relative to Earth's Primitive Upper Mantle (Righter, 2019).

(b) Mainstream gap: "The depletion of many volatile elements remains incompletely explained" (Righter, 2019).

(c) PZ prediction (Grade B; partial convergence): A protracted heliocentric phase featuring a global magma ocean would result in significant evaporative loss. This constraint is classified as active partial convergence (L-09; Grade B), reflecting qualitative consistency without a fully quantified evaporation model.

6.7 Terrestrial Geological Records (T-01, T-03 to T-08)

(a) Observed data: Records identify a global sea-level rise of ∼134 m associated with rapid deglaciation (Lambeck et al., 2014). A Carbon-14 production anomaly at ∼12,900 BP is documented in the IntCal20 calibration record (Reimer et al., 2020).

(b) Mainstream gap: The trigger for the Younger Dryas climatic reversal (∼12,800 BP) has been debated for 17 years without consensus.

(c) PZ prediction: The capture event delivered a surface energy density of 5.88×10^{25} J (equivalent to 58.8× the Chicxulub impact), consistent with the Younger Dryas proxies documented above.

6.8 Martian Constraints (Ma-01 to Ma-04)

(a) Observed data: Seismic data from InSight initially suggested a Martian core radius of ∼1,830 km (2021), later revised to $1{,}675 \pm 30$ km (2023). Over 15,000 km of ancient river systems have been mapped (Losekoot, NAM 2025 Durham).

(b) Mainstream gap: Standard climate models fail to explain the sustained internal heating required for liquid water on ancient Mars.

(c) PZ prediction: The perturbation that moved Mars from its resonant 8:5 baseline ($e = 0.30$) to its current orbit ($e = 0.09$) generated significant internal tidal heating.

6.9 Solar System Architecture (S-01 to S-05)

(a) Observed data: The 8:5 MMR between Earth and Proto-Mars (S-01) is established as the primary dynamical anchor.

(b) Mainstream gap: The precise orbital history of Mars and its anomalously low mass remain subjects of ongoing investigation.

(c) PZ prediction: S-01 is a Grade B computational result emerging from 288 WHFast simulations. Retroactive propagation of the lunar return timeline yielded a 90% credible interval for terrestrial orbit crossing of 14–117 Ma BP, overlapping with the Late Heavy Bombardment epoch — an emergent result not inserted as a premise.

7 Convergence Matrix

The evidential strength of the Paradigma Zero scenario is derived from its emergent convergence with independent observational datasets. Several of the observational constraints providing the highest discriminant power were published after the simulations were completed: Park et al. (*Nature*, May 2025), He et al. (*Nature Geoscience*, September 2025), Fischer et al. (*PNAS*, December 2024), Cai et al. (*Science Advances*, January 2025), and Barboni et al. (*Nature Communications*, March 2025). These constitute verified retrodictions R1–R5.

8 Discussion

8.1 Comparison with the Giant Impact Hypothesis

The Giant Impact hypothesis (Canup & Asphaug, 2001; Čuk & Stewart, 2012) accounts for the Earth–Moon system's angular momentum, the Moon's small iron core, and the depletion of moderately volatile elements relative to the bulk silicate Earth. The Paradigma Zero scenario does not contradict the existence of a major early impact event in Earth's history; it proposes instead that the current Moon is not the product of that event, but rather a body that formed in the same nebular reservoir as Earth — explaining the sub-ppm isotopic identity (Fischer et al., 2024) — and was subsequently captured gravitationally at ∼12,000 BP.

The principal discriminating observational differences between the two frameworks are: (1) the isotopic identity $\Delta\Delta'^{17}\mathrm{O} = 0.2 \pm 1.6$ ppm, which the Giant Impact model requires to be explained by post-impact isotopic equilibration (Pahlevan & Stevenson, 2007) or by an unusually similar impactor composition, while Paradigma Zero resolves it through co-formation; (2) the LLR kinematic paradox, which has no resolution within the Giant Impact framework given a 4.5 Gyr tidal evolution history, while a recent capture renders the paradox dynamically vacuous; and (3) the persistent lunar thermal asymmetry quantified by Park et al. (2025) and He et al. (2025), which is incompatible with 4.5 Gyr of thermal homogenization but consistent with a recent energetic event producing asymmetric tidal deformation.

The Giant Impact hypothesis does not predict a specific terrestrial obliquity value — it accounts for the existence of a large Moon that stabilizes obliquity (Laskar et al., 1993), but the specific value of 23.44° is unconstrained. In contrast, Paradigma Zero

Table 5: Cross-Convergence Matrix.

ID	Observational Data	Grade	PZ Scenario Prediction	Status
L-01	Lunar Surface Heat Flow (21/16 mW/m^2)	A	Thermal preservation from recent capture	MATCH
L-02a	Mantle Thermal Asymmetry (k_3=0.0163)	A	Asymmetric tidal pressure $r_{\min} < 20\ R_\oplus$	MATCH
L-02b	Farside Pot. Temp. (∼100°C cooler)	A	Persistent gradient from asymmetric capture	MATCH
L-03a	Liquid Outer Core (∼330 km)	A	Thermally preserved internal structure	CONSISTENT
L-03b	Solid Inner Core (258 km) + Overturn	A	Internal stir mediated by tidal capture energy	CONSISTENT
L-04	Crustal Asymmetry (30 km Near / 60 km Far)	A	$\Delta P > P_{\text{yield}}$ (silicates) during flyby	MATCH
L-05	>90% Moonquakes on Nearside	A	Asymmetric mantle stress concentration	CONSISTENT
L-06	Intense Lunar Dynamo (20–110 μT)	A	Tidal restart/intensification of core convection	CONSISTENT
L-07	K-Ar reset in Apollo samples	B	Thermal energy > closure temperature (300°C)	MATCH
L-08a	Robust U-Pb Geochronology	A	Ancient body (4.56 Ga) / Recent capture event	CONSISTENT
L-08b	Oldest preserved Zircon (4.46 Ga)	A	Consistent with pre-capture heliocentric phase	CONSISTENT
L-09	Volatile Depletion (4–5× < PUM)	B	Evaporative loss during solar wandering	CONSISTENT
L-10	Oxygen Isotope Identity (sub-ppm)	A	Co-formation in same nebular reservoir	MATCH
L-11	Pristine Clasts modification	B	Recent energetic history (∼12,000 BP)	CONSISTENT
M-01	Antipodal Magnetic Anomalies	B	Dynamo activity during post-impact cooling	CONSISTENT
M-02	Dynamo Longevity (∼2–4 μT @ 2.0 Ga)	A	Tidal heating extension of core convection	CONSISTENT
T-01	Terrestrial Obliquity (23.44°)	A	Obliquity stabilization via capture torque	MATCH
T-02	LLR Kinematic Paradox (38.08 mm/yr)	A	Extrapolation invalid due to recent capture	MATCH
T-03	Younger Dryas energy proxies	B	Energy density 58.8× Chicxulub @ 12.8k BP	CONSISTENT
T-04	IntCal20 ^{14}C Anomaly (∼12,900 BP)	B	NO_x-mediated stratospheric perturbation	CONSISTENT
T-05	Global Sea Level rise (∼134 m)	B	Obliquity-induced redistribution of insolation	CONSISTENT
T-06	YDB Nanodiamonds (10–3,700 ppb)	B	Capture pressure > 10 GPa (carbon detonation)	CONSISTENT
T-07	Bayesian synchronous age (12.8k BP)	A	Capture event temporal anchor (Event E1)	MATCH
T-08	Meltwater Pulse 1A (MWP-1A)	B	Consistent with tidal onset; $\mathrm{d}E/\mathrm{d}t \propto r^{-6}$	CONVERGE PARTIAL
Ma-01	Mars $a = 1.523679$ AU / $e = 0.09$	A	Secular g_4–g_3 migration from 1.368 AU	MATCH
Ma-02	Mars Inclination (1.85°)	A	Inclination conserved (final $i = 1.847°$)	MATCH
Ma-03a	Mars Core radius — 2021 (1,830 km)	B	Molten basal layer enhances tidal migration	CONSISTENT
Ma-03b	Mars Core radius — 2023 (1,675 ± 30 km)	A	Revised structure consistent with tidal model	CONSISTENT
Ma-04	Ancient rivers (∼3.7 Ga)	B	Tidal heating at e=0.30 (heliocentric phase)	CONSISTENT
S-01	Earth–Mars MMR 8:5 anchor	B	a=1.368 AU (Grade B computational)	MATCH (Grade B)
S-02	Jovian Irregular Satellites ($i \sim 148°$)	B	SIM3 Kozai–Lidov dynamical analog	CONSISTENT
S-03	Kirkwood Gaps (4:1, 3:1, 2:1)	B	SIM2 resonance diffusion mechanism	CONSISTENT
S-04	Late Heavy Bombardment	B	Median return time overlap (135 Myr)	CONSISTENT
S-05	Jupiter Grand Tack (4.56 Ga)	A	Temporal anchor for Phase 0 (Event E7)	CONSISTENT
S-06	Farside Seismicity Mission (2027)	—	<10% of DMQ on farside; >30% would falsify	PREDICTION

Legend: **MATCH** — quantitative prediction confirmed; **CONSISTENT** — compatible without conflict; **NEUTRAL** — non-discriminant data; **PREDICTION** — falsifiable future test. Grade A = established literature; Grade B = computational result from this work.

Table 6: Quantitative Matches (Primary Evidence).

Ref	Parameter	Observed	Predicted	Primary Source	PZ Derivation
Ma-01	Martian semi-major axis	1.523679 AU	1.524 AU	JPL Horizons	Module M2 (g_4–g_3 secular)
Ma-02	Martian inclination	1.85°	1.847°–1.848°	USNO/Pitjeva 2013	Module M1 N-body (F5-v6)
T-02	Lunar recession rate	38.08 ± 0.19 mm/yr	Paradox resolved (∼12k BP)	Williams et al. 2014	Module M4 Analysis
T-07	YDB Synchronicity	12.8k Cal BP	12,134–13,862 BP	Kennett et al. 2015	SIM5 Timeline MC
L-04	Crustal Asymmetry	2:1 ratio (60/30 km)	Match @ $r_{\min} \sim 15\ R_\oplus$	Wieczorek et al. 2013	Module M1 Capture
L-01	Surface Heat Flow	14–21 mW/m^2	∼15 mW/m^2 (@12k BP)	Langseth et al. 1976	Module M5.3 Cooling
L-10	Oxygen Isotope	0.2 ± 1.6 ppm	0.0 ppm (identity)	Fischer et al. 2024	Co-formation Axiom §3
T-01	Terrestrial Tilt	23.44°	23.5°	Laskar et al. 1993	Module M3 ($E_{\text{ratio}} = 1.0005$)

derives this value from the capture torque calculation (M3), where the energy ratio $E_{\text{torque}}/E_{\text{tilt}} = 1.0005$ across both independent candidates indicates that the scenario produces exactly the observed obliquity from first principles.

The three-timescale structure of post-capture tidal evolution (Section 5.4) provides an additional discriminant between the two frameworks. The Giant Impact hypothesis predicts continuous tidal evolution over 4.5 Gyr, producing a single monotonic recession history. The Paradigma Zero scenario predicts a fundamentally different structure: an initial high-dissipation eccentric phase ($t_{a^*} \sim 12$ ka) transitioning to a quasi-circular damping phase ($t_{e^*} \sim 57$–900 kyr) and then to a secular regime ($t_{\text{full}} \sim 1$ Gyr). This prediction is testable through high-precision tidal modeling of the lunar orbit — a gradual onset of the secular regime would be detectable as a deviation from the constant-Q tidal model assumed in standard LLR analysis.

8.2 The Role of Body Z Rejection as Methodological Control

The progressive rejection of Body Z across 936 orbital configurations constitutes an important methodological control for the Paradigma Zero result. It demonstrates that the two-body (Sun–Earth) system with Proto-Mars in MMR 8:5 is the minimal configuration consistent with both the capture geometry and the modern Solar System architecture. The absence of any viable Body Z configuration across a broad, systematic parameter sweep — not a theoretical argument, but a computational result — eliminates a class of more complex models and identifies the three-body Sun–Earth–Mars–Moon system as the parsimonious solution.

8.3 Implications for Earth–Moon–Mars System Evolution

If the Paradigma Zero scenario is confirmed, the implications for our understanding of the inner Solar System's evolution are substantial. The pre-capture Earth was in a 8:5 mean-motion resonance with Proto-Mars at $a = 1.368$ AU, $e = 0.30$ — a dynamical configuration with no analog in the present Solar System. The perturbation of this resonance by the lunar capture event provides a unified mechanism for three currently unconnected phenomena: the migration of Mars to its current orbit, the establishment of terrestrial obliquity, and the environmental disruption documented at the Younger Dryas Boundary.

The Martian tidal heating implied by the Proto-Mars eccentricity of 0.30 during its heliocentric pre-migration phase provides a candidate mechanism for the liquid water activity documented by Losekoot et al. (NAM 2025) without requiring a dense early atmosphere or specific greenhouse gas concentrations. This prediction (Ma-04) is independently testable through the geochemistry of ancient Martian hydrothermal deposits.

The terrestrial implications are equally significant. The establishment of 23.44° obliquity at ∼12,000 BP — rather than 4.56 Ga — implies that the Milankovitch cycles as currently understood are young phenomena. The redistribution of insolation following the obliquity change provides a causal mechanism for the Younger Dryas cooling episode and the subsequent Holocene warming, connecting planetary-scale dynamics to the climatic record of the last 15,000 years.

8.4 Progressive Coupling and the 15,000–11,500 BP Window

The progressive coupling framework identifies a single physical parameter — the time-varying minimum Earth–Moon distance $r_{\min}(t)$ — as the common modulator of the observed signatures in the 15,000–11,500 BP window. Because each observable scales with a different power of $r_{\min}^{-1}$ — tidal dissipation as r^{-6}, spin-orbit coupling as r^{-3}, and gravitational perturbation as r^{-2} — the model predicts a hierarchical and time-ordered response across tidal, rotational, and orbital domains. This produces a coordinated crescendo of effects across geological, biological, and astronomical records, culminating at peak coupling (∼12,800 BP, Event E1) and subsequently relaxing toward the Cassini State 2 equilibrium at ∼11,500 BP. The apparent clustering of geological and astronomical events in this window thus emerges as a deterministic consequence of the coupling dynamics rather than independent coincidences. This interpretation is consistent with the Meltwater Pulse 1A (∼14,500 BP; T-08), here reclassified from NEUTRAL to CONVERGE PARTIAL as a signal of pre-apex tidal onset, and with the D-O variability precursor documented by Brashear et al. (2025, *Climate of the Past* 21, 529–546), in which interannual climate variability leads abrupt warmings by hundreds of years — a pattern consistent with the transition from discrete flybys to continuous geocentric orbit. The progressive coupling framework is identified as a testable extension of the present model and constitutes priority future work (see Protocolo ALFA, Section 9.7, item viii).

8.5 Anticipated Objections and Responses

Objection 1 — U-Pb geochronology places the Moon at 4.46 Ga: This is fully consistent with Paradigma Zero. The Moon is ancient (forming ∼4.56 Ga BP); only the capture event is recent (∼12,000 BP). Zircon ages record the crystallization of the lunar magma ocean during the heliocentric phase, not the capture event. U-Pb in zircon is robust to thermal perturbations below 700–900°C and is not reset by the capture energy budget.

Objection 2 — Gravitational capture requires implausible energy dissipation: The dissipation requirement (Bondi, 1979) is satisfied by the Moon's global magma ocean state during its heliocentric phase ($k_2/Q = 0.400$). The peak tidal power at periapsis ($\mathrm{d}E/\mathrm{d}t|_{\mathrm{peri}} \approx 1.679 \times 10^{16}$ W) is high but not unphysical given the small periapsis distance and the highly dissipative lunar state. Triton's capture by Neptune (Agnor & Hamilton, 2006) is an accepted precedent for gravitational capture of a large satellite with analogous dynamical mechanisms.

Objection 3 — The Younger Dryas has other proposed causes: Agreed. The Paradigma Zero scenario provides a causal mechanism consistent with the YDB evidence, but does not claim to have excluded all other hypotheses. The scenario is falsifiable independently of the YDB evidence through the FSS/LITMS seismic predictions (S-06) and the K-Ar isotopic reset prediction (L-07), neither of which depends on the Younger Dryas connection.

Objection 4 — The 8:5 MMR initial condition (S-01) is Grade B computational evidence: Correct, and explicitly acknowledged in the Convergence Matrix. The theoretical framework of mean-motion resonances is Grade A (Bretagnon, 1974; Laskar, 1990); the specific 8:5 configuration is a computational result of this work and is designated accordingly.

9 Limitations and Future Work

As a report of an emergent discovery, this study recognizes several methodological boundaries and computational constraints. Declaring these limitations is essential for the scientific integrity of the Paradigma Zero scenario and defines the necessary trajectory for future validation.

9.1 Obliquity Constraint: Analytical Closure and Prospective Numerical Extension

Constraint M3 — the delivery of the angular momentum increment required to establish Earth's current obliquity of 23.44° — is analytically closed in F5-v6, with $E_{\text{torque}}/E_{\text{tilt}} = 1.0005$ and $L_{\text{Moon}}/\Delta L = 11.9$ (P1-06) and 11.3 (P1-03), verified across two independent capture geometries. The 3D SIM-3D campaign converged on candidate V31-009923 ($i = 17°$, $\omega = 286°$, $r_{\min} = 68.1\ R_{\oplus}$; three of four constraints numerically satisfied), establishing the orbital geometry consistent with M3.

These results constitute a complete analytical foundation for the obliquity constraint. The analytical closure verifies energetic consistency — that the scenario delivers the correct torque to initiate the precessional trajectory toward 23.44°. A prospective high-resolution integration campaign targeting the refined grid $a = [1.440–1.462]$ AU, $e = [0.468–0.480]$, $i = [15–21]°$ is proposed as a dedicated follow-on study providing full numerical verification.

The circularization gate mechanism (Section 5.3) constitutes the explicit physical mechanism for the long-term precessional process referenced above. The gate operates on the timescale $t_{e^*} = 57–900$ kyr (SIM-A v3), which is co-temporal with the angular momentum transfer timescale of 20–40 kyr derived analytically from the coupled-pendulum equations (Yoder, 1979). The precise final value of 23.44° is governed by the Cassini State 2 equilibrium under asymmetric dissipation ($\beta_2/\beta_1 = 6{,}215$); its exact determination requires numerical integration of the full secular system, identified here as priority future work.

9.2 Limitations of Phase 0: The Heliocentric Phase

The simulations modeling the pre-capture heliocentric history (SIM1–SIM5) incorporate several simplifications that affect the timing of intermediate events without altering the underlying physical mechanisms:

(a) 2D dimensionality (±20–40% timing uncertainty);

(b) numerical precision for encounters $d < 0.001$ AU;

(c) EKH tidal terms applied exclusively to the Moon;

(d) static Martian baseline at 1.368 AU;

(e) $N = 36$ in SIM2 (insufficient for Bayesian credible intervals below 10%);

(f) chaotic diffusion rate σ_e uncertainty of ∼3× when applied to a lunar-mass body;

(g) irreducible degree of freedom E4b (340 Ma to 4.56 Ga BP);

(h) unmodeled gravitational interactions with the Galilean satellites during the Kozai phase.

9.3 Reconciliation with U-Pb Geochronology

High-precision U-Pb dating of lunar zircons remains robust (L-08a, L-08b), with the oldest preserved clasts yielding ages of 4.460 ± 0.031 Ga (Greer et al., 2023). The Paradigma Zero scenario resolves the apparent chronological conflict by establishing that the Moon is ancient (forming $\sim$4.56 Ga BP), while the gravitational capture is a recent geodynamic event ($\sim$12,000 BP).

9.4 Lunar Volatile Depletion (L-09)

The Moon is significantly depleted in volatile elements (Na, K, Rb, Cs) compared to Earth's Primitive Upper Mantle, with depletion factors of 4–5 measured by Righter (2019). While the scenario proposes that this depletion resulted from evaporative loss during a protracted heliocentric phase, a formal quantitative derivation is pending and remains a subject for future investigation.

9.5 Status of the 8:5 Mean Motion Resonance (S-01)

The identification of the 8:5 Earth–Mars MMR as the study's dynamical anchor is classified as Grade B computational evidence, as it emerged from the study's own WHFast integrations (S-01). However, the underlying theoretical framework of MMR and secular stability (Bretagnon, 1974; Laskar, 1990) constitutes Grade A evidence from established planetary science. Maintaining this distinction between computational discovery and theoretical basis is paramount for the transparency of the model.

9.6 Sensitivity to Tidal Dissipation Parameters

The adopted tidal dissipation parameter $k_2/Q = 0.400$ represents the fluid-body upper limit for a lunar-sized body in the global magma ocean state. The sensitivity of the capture outcome to this parameter has not been systematically explored in the present work. A parametric sweep of k_2/Q over the range $[0.01, 0.50]$ is identified as a necessary component of future validation.

9.7 Future Work

The results presented in this study open several well-defined research directions:

(i) **High-resolution 3D obliquity extension:** prospective integration campaign over the refined grid $a = [1.440–1.462]$ AU, $e = [0.468–0.480]$, $i = [15–21]^\circ$.

(ii) **High-inclination heliocentric re-integration:** re-execution of SIM1–SIM5 with real orbital inclinations and IAS15 for sub-0.001 AU encounters.

(iii) **Statistical expansion of SIM2:** increase from $N = 36$ to $N \geq 1{,}000$ for proper Bayesian credible interval construction.

(iv) **Full EKH dissipation model:** extension of tidal dissipation terms to the young Earth and Proto-Mars.

(v) **Quantitative volatile evaporation model:** development of a mass-loss model for Na, K, Rb, and Cs during the magma ocean phase.

(vi) k_2/Q **parametric sensitivity sweep:** systematic exploration of capture outcome over $k_2/Q \in [0.01, 0.50]$.

(vii) **Variable-dissipation tidal model — SIM-A v2/v3 (Limitation L3, RESOLVED FOR P1-03):** SIM-A v2 (120 runs) resolves L3 for P1-03: 19 configurations produce t_{a^*} consistent with $E_0 = 12{,}000$ BP (best: 12.19 kyr, ratio 1.02×). SIM-A v3 (275 total runs, of which 100 for P1-03 are reported in Section 5.4.2) establishes the three-timescale structure: $t_{a^*} = 12$–100 kyr; $t_{e^*} = 57$–900 kyr; $t_{\text{full}} \sim 1$ Gyr. Remaining work: (a) k_2/Q parametric sweep for P1-06 (extended apoapsis grid); (b) validation against REBOUNDx `tides_spin`.

(viii) **Protocolo ALFA — Isotopic variability as a dynamical response to progressive orbital coupling.** The progressive coupling framework (Section 8.4) identifies $r_{\min}(t)$ as the primary control parameter governing Earth–Moon interaction strength during the pre-capture phase. Spin-orbit coupling scales as $r_{\min}^{-3}$; tidal dissipation power scales as $r_{\min}^{-6}$.

We propose that isotopic variability in high-resolution ice core records does not directly track the absolute magnitude of energy input, but rather the temporal modulation of the effective coupling strength. Specifically, the observed variance responds primarily to the temporal variability and irregularity of the orbital forcing — to the rate of change and discontinuity structure of the coupling modulation rather than to its mean amplitude.

Three orbital regimes emerge naturally: (1) during the pre-capture flyby phase, repeated close encounters produce large, irregular fluctuations in $r_{\min}(t)$, yielding elevated isotopic variance; (2) as the system transitions to a bound orbit, variability of $r_{\min}(t)$ decreases, leading to a reduction in high-frequency isotopic variance despite continued energy input; (3) following capture, continuous tidal dissipation produces smooth coupling and low variance. This interpretation is consistent with Brashear et al. (2025), which documents a reduction in interannual climate variability preceding D-O warming transitions.

This framework yields three falsifiable predictions testable against publicly available ice core data (WAIS Divide, NGRIP, EGRIP; PANGAEA doi:10.1594/PANGAEA.902577):

(P1) A measurable decline in the rolling variance of δ^{18}O in the 7–15-year band during the centuries immediately preceding the YDB (∼12,800 BP), consistent with the transition from discrete flyby forcing to continuous orbital coupling.

(P2) The same two-regime variance structure independently present in at least two of three ice core records (WAIS Divide, NGRIP, EGRIP), with regional phase offsets but qualitatively consistent morphology.

(P3) A non-monotonic variance trajectory over the 25,000–12,800 BP window, characterized by intermittent variance peaks consistent with discrete forcing events rather than a smooth monotonic trend.

Confirmation criterion: detection of a pre-event variance decline (P1) with onset within 13,500–12,800 BP and duration of 200–1,000 years, present in ≥ 2 independent cores. *Falsification criterion:* monotonically increasing or stationary variance in all three cores throughout the 15,000–12,800 BP window, with no pre-event decline detectable above the 2σ noise floor. *Cost:* \$0. *Estimated execution time:* 2–5 days.

10 Conclusions

This work presents computational and observational evidence for a dynamically consistent scenario of gravitational capture of the Moon by the Earth, designated the Paradigma Zero scenario. The principal findings are as follows.

A systematic unguided parametric exploration of over 65,000 N-body integrations, executed with REBOUND/IAS15 across 26 simulation stages, produced two canonical capture candidates — P1-06 ($r_{\min} = 15.51\ R_\oplus$, $\Delta T = 17.8$ yr) and P1-03 ($r_{\min} = 9.84\ R_\oplus$, $\Delta T = 122.6$ yr) — after systematically rejecting a more complex third-perturber hypothesis (Body Z) that was falsified by the simulations themselves. The rejection of Body Z is a computational result, not a theoretical assumption.

Both candidates satisfy the orbital viability constraints M1–M3 with high fidelity: Earth's orbital stability ($a_f = 0.9996$–1.0004 AU); Proto-Mars secular migration to its present orbit via the g_4–g_3 resonance with Jupiter in $\tau_e \approx 1.0$ Myr ($a_{\rm PM}^{\rm final} = 1.4954$–$1.5169$ AU); gravitational capture within $r_{\min} < 70\ R_\oplus$. The energetic consistency of terrestrial obliquity establishment yields $E_{\rm torque}/E_{\rm tilt} = 1.0005$ independently for both candidates, suggesting a fundamental constraint of the scenario rather than parameter tuning. The LLR kinematic paradox (M4) is qualitatively resolved by the recent capture hypothesis; quantitative consistency with the chronological anchor E_0 requires a variable-dissipation tidal model for the eccentric post-capture phase, identified as Limitation L3 and high-priority future work.

Post-hoc evaluation against 32 independent peer-reviewed observational datasets produces no contradiction. Five verified retrodictions (R1–R5) are documented, all published after the simulations were completed: R1 (Park et al., 2025), R2 (He et al., 2025), R3 (Fischer et al., 2024), R4 (Cai et al., 2025), and R5 (Barboni et al., 2025). Eight quantitative matches between simulation-derived predictions and independent observations are reported in Table 7.2.

Two formally registered falsifiable predictions provide the objective means for empirical determination. Prediction S-06 specifies that fewer than 10% of deep moonquakes should occur on the farside, testable by the FSS+LITMS dual instrument system at Schrödinger Basin (launch 2027; confirmation criterion <10%, falsification criterion >30%). Prediction L-07 specifies K-Ar isotopic reset in Apollo samples from tsunami-impacted zones at 12,000–13,000 BP, with U-Pb in zircon remaining unchanged as an independent chronological anchor.

The scenario presented here identifies a previously unexplored dynamical regime of Earth–Moon encounters and demonstrates that post-capture orbital configurations consistent with present-day constraints can be constructed. SIM-A v3 reveals that post-capture tidal evolution is governed by three physically distinct timescales: $t_{a^*} \sim 12$ ka (semi-major axis reaches present value, anchoring E_0); $t_{e^*} \sim 57$–900 kyr (eccentricity damps to present value, secular continuation); $t_{\rm full} \sim 1$ Gyr (simultaneous modern configuration, LLR-observable secular regime). The chronological anchor

$E_0 = 12{,}000$ BP is associated with t_{a^*}; the LLR secular rate reflects t_{full}. LLR data and geological records are both compatible with the hypothesis of recent capture when considered within their respective domains and timescales. Whether the Paradigma Zero scenario ultimately proves to be the correct historical description of the Earth-Moon system will be determined by empirical data. The testable predictions established for the 2027 seismic and heat flow missions provide the objective means for that determination.

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